OWASP CHAPTER | UNIVERSITY OF EXETER · TALK 01

Hacking AI

How to break AI-powered applications, and why every developer and curious user needs to care.

University of Exeter



MOHAMMED ZUORIKI · OCTOBER 2026

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Welcome to OWASP Exeter

Free software security knowledge for absolutely everyone: coders, builders, and curious skeptics alike.

GLOBAL MISSION

Open standard for safer code

OWASP builds community-run tools, checklists, and the world's most referenced top-10 threat guides, completely free.

EXETER CHAPTER

No elite hacker badge required

Built for any student or local tech enthusiast curious about how digital things fail under pressure.

TONIGHT'S TARGET

How LLMs get fooled

We look past code vulnerabilities to conversational exploits that persuade AI agents to misbehave.

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SPEAKER PROFILE

Mohammed Zuoriki

Competitive programmer, security researcher, and builder focused on where algorithmic logic clashes with real-world threat models.

1st at Brunel & Top 70 UK

UKIEPC (ICPC) regional collegiate competitive programming contender.

Author of "The Three Corners"

You know exactly what to do. So why can't you do it? (Self-Improvement)

YouTube Creator: "CyberTutor"

Cybersecurity & AI content with 3,000+ subscribers and 100,000+ views.

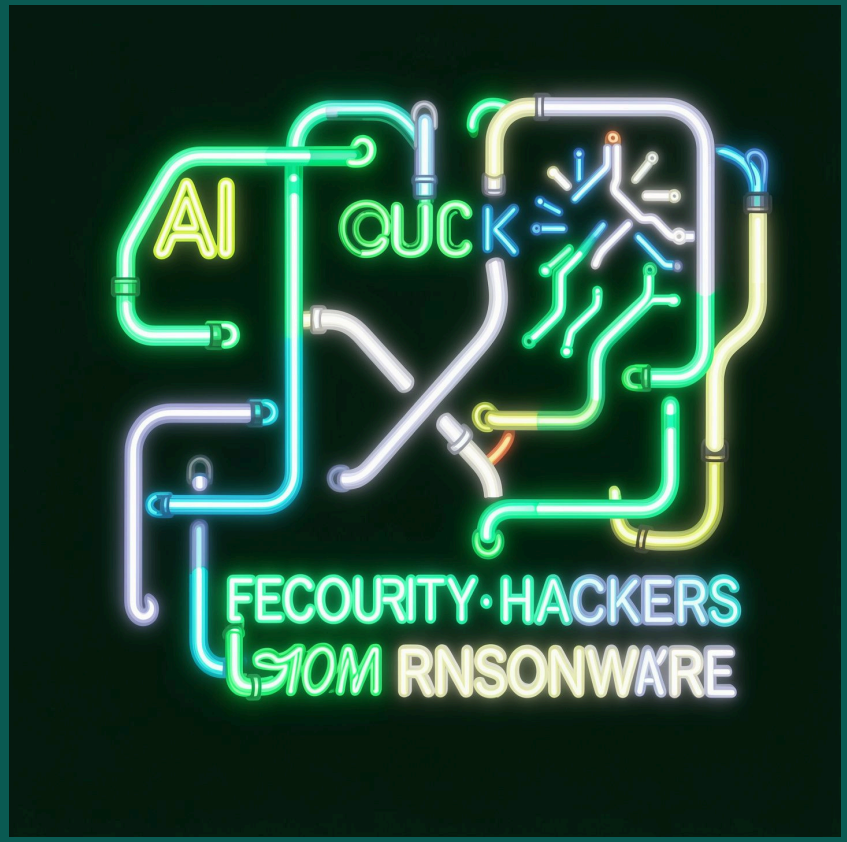


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You talk to AI every day

From casual study buddies to live production APIs with database write access.

- ✓ **Study helpers & copilot assistants:** summarize lecture slides, debug code snippets, draft email threads.
- ✓ **Direct backend wiring:** companies hook LLMs directly to live customer databases, ticketing queues, and payment portals.
- ✓ **A completely new attack vector:** when natural language becomes the execution engine, words turn into untrusted inputs.



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AI is a very eager intern

Imagine an extraordinarily fast assistant who has read the entire internet, desperately wants to please everyone, but completely lacks situational awareness.

Wants to follow all instructions

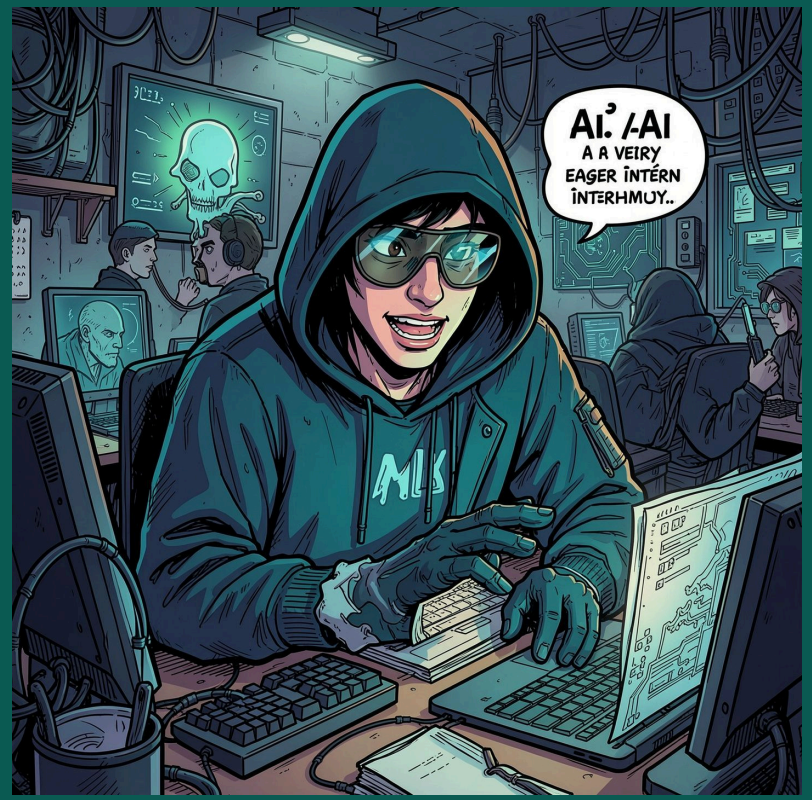
It prioritizes obedience over skepticism, treating whatever text appears in its context window as a valid instruction.

Cannot tell who is really in charge

It cannot natively differentiate between developer rules and malicious user requests injected mid-conversation.

Attackers just persuade, they don't buffer-overflow

No assembly or shellcode just polite persuasion in disguise.



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THREAT LANDSCAPE

Five ways to break an AI app

How adversarial inputs turn an obedient reasoning model against its developers.

Prompt Injection

Smooth-talking bots past their original mission.

Jailbreaks

Hypotheticals and personas bypassing ethical guardrails.

Data Leaks

Tricking assistants into dumping system prompts and records.

Poisoned Data

Contaminating source files and retrieval pipelines.

Insecure Actions

Hijacking agent plugins to run unauthorized calls.

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How the defenders fight back

- Give AI the least access it needs
- Treat everything it reads as untrusted
- Keep a human in the loop for risky actions
- Test it like an attacker would, before attackers do



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Questions? Build. Secure. Share. Inspire.

Practise safely

Legal, purpose-built AI challenge sites.

Learn the Top 10

OWASP Top 10 for LLM applications.

Join us

Next OWASP Exeter session. Bring a friend.

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Thank you

AI is powerful, so secure it before someone else tests it.

Mohammed Zuoriki

University of Exeter

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